

CA-2 - Research Methodology - Literature review - Synthetic Dataset Generation for UAV Detection and Tracking

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1 Related work

NOTE: This has been titled as "Related Work", since the idea is to try to publish my implementation by the end of the semester. Kindly consider.

There is a clear gap between synthetic and real datasets, even beyond just the context of UAV detection - the primary operating domain of the trained models is in the real world. Training a model exclusively on synthetic data introduces a domain gap between the synthetic training data and real world conditions in which the model is expected to operate. Hence in recent years, to minimize this gap, research in synthetic dataset creation has been exploring domain randomization and adjacent topics.

1.1 Synthetic Data and Domain Randomization

Synthetic data generation addresses the issue of sparse availability of clear, well annotated datasets. It also allows for capture of difficult to replicate practical conditions in data. However, it also faces the challenge of being highly deterministic. So, to mitigate this, SDR is utilized, while DR is optionally used to augment each element of the dataset.

Domain Randomization, Structured Domain Randomization, Guided Domain randomization, etc. are needed to ensure that the created dataset is diversified.

Domain Randomization (DR) is used to introduce random variations (textures, distractors, etc.)

into synthetic data, allowing models to generalize across a set of diverse scenarios, including unseen ones. Giving this approach a little more context to work with, Structured Domain Randomization (SDR) places these variations into more realistic scenarios - like birds adjacent to drones.

According to Zhu et al. [1] DR can operate across five components - object randomization, background and distractor randomization, camera randomization, illumination randomization, and post-processing.

Maddikunta et al. [2] demonstrate a broader application of domain randomization for document recognition in natural scenes. Their approach randomizes scene factors such as camera position and orientation, focus, background, document geometry, illumination, occlusion and cropping. They additionally randomize document content and style to reduce the dependence of the trained model on specific textual content or physical document appearance. This demonstrates that domain randomization can be extended beyond the visual properties of the scene itself to control the information contained within the target object.

Wisniewski et al. [3] limit the scope of DR utilized in their approach. They employ structured domain randomization and domain randomization of variables such as camera focal length and camera position to improve the generalizability of the synthetic training data.

Another interesting point - Zanardelli et al. [4] render an abstract outdoor scene containing diverse objects with different materials and reflectance properties. Lighting is varied by pushing the sun through one solar cycle. They generate ground-truths such

as geometric and semantic annotations, instance-segmentation masks and per-pixel surface normal maps. This explicitly demonstrates that the benefits of synthetic data extend into generating otherwise difficult to obtain types of data, while reducing the manual error involved in defining the ground truth.

On the other hand, Lenhard et al. [5] utilize SDR in order to generate a synthetic dataset - they use a defined set of environments (and lighting conditions) along a defined set of drone 3D models. With these pieces constant, their combinations are varied to capture RGB images from different points-of-view.

Finally, Barisic et al. [6] propose an interesting idea - the idea of texture-invariance by using DR to randomize the textures for each object of interest in the image.

1.2 Geometry versus Texture (Visual Cues)

Before getting to texture invariance, Zhu et al. [1] demonstrate the utility of texture. They look at a different context - manufacturing. Unlike approaches that rely on broad randomization, the study systematically evaluates these factors across a public robotic dataset and the newly introduced SIP15-OD dataset, containing 15 industrial objects across three manufacturing use cases. The carried out ablation studies show that importance of texture depends strongly on the imaging context and physical characteristics of the target objects.

Now, Barisic et al. [6] point out an interesting distinction. The study focuses on two challenging conditions in aerial imagery: varying illumination and the small relative size of UAVs in images. Therefore, at significant distances between the object of interest and the camera, shape-oriented detection is a more robust approach.

Thus, the usefulness of texture as a visual cue appears to depend on the imaging regime: close-range objects may retain informative material and texture cues, whereas distant UAVs may provide insufficient spatial resolution for texture to remain reliable.

Even in this shape-oriented approach, similarly shaped distractor objects give false positives, which

can be tackled by including negatively-labeled similar shape objects in the dataset.

1.3 Sim-to-Real Transfer Challenges

Sim-to-real transfer is a core point that this field of research is trying to tackle. Generalization of models trained on synthetic data face issues while dealing with underexposed imagery, difference in lighting conditions, false positives, and so on.

Zhu et al. [1], based on their results, state that even with the use of extensive DR, inclusion of post-processing in the dataset such as noise and blur further increase the robustness of the model trained on it.

Wisniewski et al. [3] approach sim-to-real transfer with great detail. They tested two main factors - distractors and camera bounds.

1.3.1 Distractors

They created five variations of their datasets - Drones Only, Drones with Birds, Drones with Generic Distractors, Drones with Realistic Distractors and Drones with Unrealistic Backgrounds - to observe if any of these dataset variations, when used to train their detection model, would result in improvements.

The result here challenges the assumption that distractors would simply improve sim-to-real transfer. They state, that the drones-only, drones-and-birds, and generic-distractor datasets performed similarly on MAV-Vid: 97.0%, 97.3%, and 97.3% AP@0.5, respectively. On Drone-vs-Bird, however, the drones-only dataset performed best at 49.7%, while generic distractors achieved 48.2% and, surprisingly, adding synthetic birds reduced performance to 40.3%. The authors therefore found no significant improvement from the tested domain-randomization styles over the basic drones-only dataset. Therefore, simply reproducing a real-world distractor is not sufficient to improve sim-to-real transfer; the characteristics and distribution of the distractors may matter as much as their presence. [3]

1.3.2 Camera bounds

Again, they created five variations of their datasets, each with different camera bounds : 20m - 40m - 80m - 160m - 320m. The results show that 20–80 m produced relatively similar performance, with AP@0.5 - MAV-Vid: 95.0–97.0%; Drone-vs-Bird: 49.8–53.2%; Anti-UAV: 66.3–67.8%. At 160 m, MAV-Vid performance dropped substantially to 88.0%, although Drone-vs-Bird remained competitively at 53.6%. Finally at 320 m, performance dropped significantly across all three datasets because the model struggled to learn the drone’s shape at such small scales.

The result here supports an inference from the previous section (Geometry versus Texture) that relative position of the object-of-interest with respect to the camera changes what visual information is available to the detector. At moderate distances, increasing the camera bounds can improve transfer to datasets containing smaller UAVs. But beyond a point, the UAV becomes so small that the network struggles to learn its shape. [3]

2 Conclusion

An apt conclusion would be that the effectiveness of synthetic-data randomization is not universal; it depends on what variation actually exists in the target deployment environment and what visual information remains available to the detector.

As observed from the paper by Wisniewski et al. [3], it is not enough for a variation to exist in the real world; its usefulness depends on whether that variation affects the visual cues used by the detector.

References

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